

$n \geq 30$. We could have thought of Poisson but since
 $\therefore$ We are going to assume Gaussian.

Cookies:

X_i are iid $N(\mu, \sigma^2)$.

I am going to assume σ^2 is known
and it is

$$\sigma^2 = (4.37)^2. \quad \bar{X}_n = 14.77.$$

Now, I want to perform a test
about μ .

$$H_0: \mu \geq 20. \quad \text{vs } H_1: \mu < 20.$$

Status Quo.

$\uparrow$
(this was established)
by printing on the box.

It could be, that nothing was printed
on the box and some students went
to the packaging company and said

you should print less than 15 on your box.

They would say, let's perform a status quo, you just established and it's not quite correct. $\rightarrow H: \mu_0 < 15$.

$\bar{X}_n = 14.77$
when should I reject,

I don't know, what distribution of $\bar{X}_n$ actually is, But for $n > 30$, the test t-test or what we are going to use are the same.

We should use something called a t-test.

In gen. it should think of this

$$\left(\frac{\sqrt{n} (\bar{X}_n - \mu)}{\sigma} \right)$$

It is an estimate of std. deviation. we are not there yet.

Here, by Slutsky

No. of chocolate chip/cookie is binomial and by CLT Normal

$$\sup_{\mu \geq 20} P_{\mu} \left[\sqrt{n} \left(\frac{\bar{X}_n - \mu}{s} \right) < -q_{\alpha} \right] = \alpha.$$

uniformly over all μ 's that are at least 20.

Sup is achieved at boundary $\mu = 20$.

Therefore

$$R_{\psi} = \left\{ \sqrt{n} \frac{\bar{X}_n - 20}{4.37} < -q_{\alpha} \right\} = \mathcal{R}$$

$$\sqrt{32} \left(\frac{14.77 - 20}{4.37} \right) = -6.77.$$

p -value $< 10^{-4}$.

we reject.
 $\therefore$ what's written on the box is wrong and they should change it.

sup is achieved at boundary = 20.

$$\therefore \left\{ \frac{\sqrt{n} \bar{X}_n - 20}{\sqrt{2.5}} < -z_{\alpha} \right\}$$

$$\downarrow$$

$$-6.77$$

$$P(|Z| < z)$$

$$P(|Z| < z)$$

$$2\phi(z)$$

$$P(Z < z) + P(-Z < z)$$

$$-6.77 < -z_{\alpha}$$

$$\downarrow$$

$$< 10^{-4}$$

we reject

$$P(|Z| < z_{\alpha/2}) = 1 - \alpha$$

$$= P(Z < -z)$$

$$1 -$$

$$2\phi(z)$$

$$1 - 2\phi(z) > \alpha$$

$$2\phi(z) - 1 = \alpha_{2.5}$$

$$2\phi(z) = \alpha_{2.5}$$

$$\sqrt{200} \cdot 1.2 > 2.83.$$

$$z_{\alpha/2}$$

$$1.96.$$

$$z_{\alpha/2} = 1.96.$$

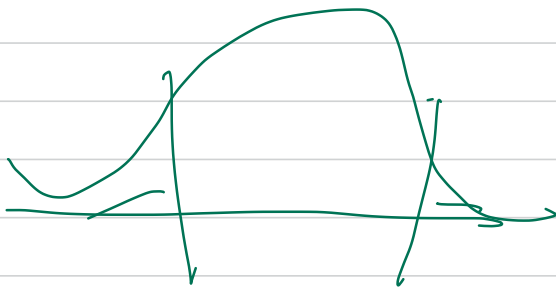
$$n_{0.05}$$

$$P(|Z| < x)$$

$$= P(Z < x) + P(Z > -x)$$

$$= 2P(Z < x) = 2\Phi(x)$$

$$\Rightarrow P(|Z| > z_{\alpha/2}) = \alpha.$$



$$P(|Z| > z_{\alpha/2})$$

$$= 1 - \Phi(x) - \Phi(-x)$$

$$P(Z > |x|)$$

$$1 - \Phi(|x|).$$

6.2 Statistics, models and Assumptions;

→ whole Trinity!

① Estimation

② Confidence Intervals

③ Hypothesis Testing.

Model and Assumptions;

Example: Boarding a flight!

HYPOTHESIS TESTING: We will ask a very simple question based on data, is method no.1 better than method number 2?

What is the fastest method to board a plain (R2F) or (Window, Middle Aisle)?

↳ It is basically from an outside to inside method.

↓
In this, board the window passengers first, then middle, then aisle.

As τ shrink down to 0, I always reject.

p-value is not the value of your test statistic.

p-value is not a no. which is 0 or 1.

No matter whether you're using asymptotic or non-asymptotic).

$$P\left(\frac{\bar{X}_n - \frac{1}{2}}{\sqrt{n}} > 3.229\right) = \text{p-value}$$

$$P(|Z| > 3.229) < \underline{10^{-4}} = \alpha$$

$\hookrightarrow Z \rightarrow N(0,1)$

For 2-sided, it will be 2 times
Area under the
curve
for > 3.229 and < -3.229
else > 3.229

Calculating the p-value.

- ① Build the statistic $\rightarrow$ Decide and α .
 $\hookrightarrow$ Standardized Count to Gaussian
- ② Plug in the numbers $= X$
- ③ Compute $P(|Z| > X)$
 $\downarrow$
 $P(N(0,1) > X)$.

Hypothesis Testing

Recitation!

→ You-Shop-owner- have one employee.

says I should not
come (at 10 AM) every morning
i.e. there is only 2 customers.

You decide to test whether
to keep the shop open or not.

$X_1, \dots, X_n \overset{iid}{\sim} \text{Pois}(\lambda), \lambda > 0$

→ you decide to

Goal: Find test $\varphi(x_1, \dots, x_n)$ test for n -days
with asymptotic level α .

(1) $H_0: \lambda = 2$

$H_1: \lambda \neq 2$

$H_0 = \{2\}$

Type I error: $\alpha_\varphi(\lambda) = P_\lambda$
 H_0 is true when it's not. $\lambda \in \mathbb{R}_0^+$

$\alpha_\varphi(\lambda) = P_\lambda(\varphi(X_1, \dots, X_n) = 1), \lambda \in \mathbb{R}_0^+$

Type II error: $\beta_\varphi(\lambda) = P_\lambda(\varphi(X_1, \dots, X_n) = 0)$
Null hypothesis is true when it's not. $\lambda \in \mathbb{R}_0^+ \setminus H_0$

→ we want β to be small and we want the Power to be large.

$$\lim_{n \rightarrow \infty} \sup_{\lambda \in H_0} P_{\lambda}(\psi(X_1, \dots, X_n) = 1) = \alpha.$$

Level: $\sup_{\lambda \in H_0}$
(Asymptotic).

only reject H_0 , never accept it.

(a) Fi

$$\psi = I(T_n > S), \quad S = \frac{\ln(\lambda - 2)}{\lambda}$$

$$(b) P_{\lambda} \left(\frac{\sqrt{n} \left(\frac{\ln(\lambda - 2)}{\lambda} - \frac{\ln(\lambda - 2)}{\lambda} \right)}{\sqrt{\frac{1}{\lambda^2}}} > q_{\alpha} \right) = \alpha, \quad S = q_{\alpha}$$

$$P_2 \left(\left(\sqrt{n} \left| \frac{\ln(\lambda - 2)}{\lambda^2} \right| \right) > S \right) \xrightarrow{n \rightarrow \infty} P(|Z| > S)$$

$$\Rightarrow P(S) = 1 - \alpha, \quad S = q_{\alpha/2}, \quad \text{quantile of } N(0,1)$$

LECTURE 8:

$$3n^2 \times 1^2 = 1^2(\mu)$$
$$3n^2 \times 1^2 = 1^2(\mu)$$

UNIT 8: METHODS OF ESTIMATION:

→ Principled ways of building an estimator.
Previous:

→ Some proportion.

→ Some expected value.

KISS EXAMPLE:

→ Average of some indicators. .) →
(LLN told us it was a good idea)

we used
used $\bar{x}_n$

τ -example:

$$\left(\frac{1}{\lambda} \rightarrow \frac{1}{\bar{\lambda}_n} \right)$$

→ METHOD-OF-MOMENTS:

↳ Generic easy to use

→ when we start to have more parameters.

$(\theta \rightarrow d \text{ for } d)$?

↳ At least 2.

→ We want to estimate both μ and σ^2 .

↓
Again easy

METHOD OF MOMENTS gives you
an easy way to do this.

Much more powerful method.

MAXIMUM LIKELIHOOD ESTIMATOR:
→ Not just by computing μ and σ^2 .

↙ Principled ↘
↗ Generic!

→ your statistical model has nice properties

↙
(Enjoy some Asymptotic properties)

→ Asymptotic Normality $\downarrow$ (Best variance)

→ Variance → Asymptotic Variance.



(OPTIMALITY) → MLE.



Computing a maximum,

(External Algorithms)



Maximum likelihood

estimation,

M-ESTIMATORS:

→ Closer to machine learning.

→ Not about trinity of statistical inference

→ They are about prediction.

→ Not trying to maximize likelihood but precision.

→ Put IN THE IDEA OF WHAT
PRECISE PREDICTIONS-ACCURACY
ARE.

8.4. INTRODUCTION TO TOTAL VARIATION DISTANCE.

→ Definition:

INTUITION:

→ when building statistical method,
we know our goal



ESTIMATE OUR DISTRIBUTION.

→ If my parameter is identified



ESTIMATE THE PARAMETER.

If



I do this → I find the distribution



But my probability is close to
 true probability $\rightarrow$ by $P[\hat{P}_n - P > \epsilon] \xrightarrow{n \rightarrow \infty} 0$
 $\downarrow$
 large.

So distortion.

$$\theta \rightarrow \theta^n$$

this is
a def'n

$\theta \rightarrow$ true parameter.

$$P_{\theta^n}(A) - P_{\theta}(A)$$



$$2 \quad P_{\theta}(A) - P_{\theta^n}(A)$$

If want $\hat{\theta}$ to give me a distribution close to θ^* .



WHAT IS MY GOAL?



If my model is not identifiable,
↓ Does not matter.

I JUST WANT THE TWO DISTRIBUTIONS TO BE CLOSE.



And $\hat{\theta}_1$ vs θ^* could be different distributions.

→ what does it mean for 2 probability distributions to be close?

↓ Imp:

Ans: when I compute the probability,
I want to get pretty much the same
result.

TOTAL VARIATION DISTANCE:

one does not have to be the true one,
and one does not have to be coming from
your estimator.

$$TV(P_\theta, P_{\theta'}) = \max_{A \in \mathcal{E}} |P_\theta(A) - P_{\theta'}(A)|.$$

↑

↑ worst difference
in probability

Worst deviation when I compute those
two events under $P_\theta, P_{\theta'}$ respectively -

over all possible
events

Those are measurable subsets of my sample
space.

which is saying those
same 2 are the same

↓
equal to zero, the max is
equal to 0, means all those
numbers are equal too.

Compute Total variation
distance between those 2,

↓
Hard $\therefore$ you would
have to maximize over all subsets.

↓
A lot of combinations

→ Nice Analytical Defⁿ for TV distance:

↓
Different for
PDF AND PMF

0

1.

Theorem



If it is a Bernoulli
of 1, compute total
distance
and divide by 2.

$$\frac{1}{2} (p-q)$$
$$\frac{1}{2} - \frac{1}{3}$$

$$\frac{1}{2}$$

$$\frac{1}{2} \cdot \frac{1}{3} \leq \frac{1}{6} \quad \times$$

$$\int dx.$$

Algebra.

x

→

$$TV(P_\theta, P_{\theta'}) = \max_{A \in \mathcal{E}} |P_\theta(A) - P_{\theta'}(A)|.$$

← This is the maximum distance over all possible events of $|P_\theta(A) - P_{\theta'}(A)|$.

max is at least ϵ
that no. ϵ is found one
which is at least
equal to that number

$$= \frac{1}{2} \sum_{x \in \mathcal{E}} |P_\theta(x) - P_{\theta'}(x)|.$$

$$TV(P_\theta, P_{\theta'}) = \max_{A \subseteq E} |P_\theta(A) - P_{\theta'}(A)|.$$

$$\rightarrow |P_\theta(A) - P_{\theta'}(A)| = \frac{1}{2} \sum_{x \in E} |P_\theta(x) - P_{\theta'}(x)|.$$

There exists A such that,

$$TV(P_\theta, P_{\theta'}) = \max_{A \subseteq E} |P_\theta(A) - P_{\theta'}(A)| \rightarrow \text{is actually}$$

equal to that number.

$$= \frac{1}{2} \sum_{x \in E} |P_\theta(x) - P_{\theta'}(x)|.$$

$$\frac{1}{8} - \frac{2}{8} + \frac{1}{8}$$

is actually equal to (the largest no. it can take) i.e.

We know, there is one just A and it's $C \subseteq E$

So, it's a set $x \in E$

I want such an x .

$$A = \{x \in E: P_\theta(x) \geq P_{\theta'}(x)\}$$

$$|P_\theta(A) - P_{\theta'}(A)| = \left| \sum_{\substack{x: P_\theta(x) \geq P_{\theta'}(x)}} (P_\theta(x) - P_{\theta'}(x)) \right| \quad \text{--- (1)}$$

we can do this (11)

$$\begin{aligned} \text{i.e.} \quad & \sum P_\theta(x) - \sum P_{\theta'}(x) \\ & \quad \downarrow \quad \quad \downarrow \\ & P_\theta(A) \quad P_{\theta'}(A) \end{aligned}$$

Since these no.'s are non negative

(SUMMING OVER A^c)

$$\sum_{x: P_\theta(x) < P_{\theta'}(x)} (P_{\theta'}(x) - P_\theta(x)) = P_{\theta'}(A^c) - P_\theta(A^c)$$

(2)

$$\text{i.e.} \quad \sum_{x: P_\theta(x) < P_{\theta'}(x)} (1 - P_\theta(x)) - (1 - P_{\theta'}(x)) = 1 - P_\theta(A^c) - 1 + P_{\theta'}(A^c)$$

Summing over A^c

$$\begin{aligned} \therefore \sum_{x: P_\theta(x) < P_{\theta'}(x)} (P_\theta(x) - P_{\theta'}(x)) &= P_\theta(A) - P_{\theta'}(A) \\ &= \sum_{x: P_\theta(x) < P_{\theta'}(x)} (P_\theta(x) - P_{\theta'}(x)) \end{aligned}$$

$\therefore$ If I sum the 2 inequalities,

$$\textcircled{1} + \textcircled{2} \quad \checkmark$$

$$\therefore \sum_{x \in E} |p_\theta(x) - p_{\theta'}(x)| = 2 |p_\theta(A) - p_{\theta'}(A)|$$

$$\max_{A \subseteq E} \therefore |p_\theta(A) - p_{\theta'}(A)| = \frac{1}{2} \sum_{x \in E} |p_\theta(x) - p_{\theta'}(x)|$$

TV between $(\text{Exp}(1), \text{Unif}(0,1))$.

$$f(x) = e^{-x} \mathbb{I}(x > 0) \quad g(x) = 1 \quad (0 < x < 1)$$

$$\frac{1}{2} \int |e^{-x} \mathbb{I}(x > 0) - \mathbb{I}(0 < x < 1)|$$

$$= \int_0^1 |e^{-x} - 1| dx + \int_1^\infty e^{-x} dx$$

$$= \frac{1}{2} \left(\int_0^1 (1 - e^{-x}) dx + \int_1^\infty e^{-x} dx \right)$$

$$= \frac{1}{2} \left[(-e^{-x})_0^1 - \frac{1}{1} + \left(\frac{1}{e} - 1 \right) \right] = \frac{1}{2}$$

$$|a-b| = |a-b|$$

$$\int_0^1 P(\theta', \theta) = \tau v(\theta', \theta)$$

$$\tau v(\theta, \theta') \leq 1$$

$$\theta = \theta'$$

$$\tau v(\theta, \theta'') + P(\theta'', \theta')$$

distance

$$= \frac{1}{2} \left(-1 + \frac{1}{e} + 1 + 0, \frac{1}{e} \right)$$

$$= \frac{1}{2} \frac{2}{e} = \frac{1}{e}$$

1 2.8.

$$TV(P, Q) = M. = 1.$$

$$e^{-x} \quad v(0,1).$$

Writing $(X, X+a)$ in terms of I.R.V.

$$X = \begin{cases} 0 & \text{w.p. } 1/2 \\ 1 & \text{w.p. } 1/2 \end{cases} \xrightarrow{\text{I.R.V.}} X+a = \begin{cases} a & \text{w.p. } 1/2 \\ 1+a & \text{w.p. } 1/2 \end{cases}$$

The two R.V.s. have disjoint support.

We have to sum over 4 terms
 $\text{coz } a \in [0,1).$

→ a is not taking values 0 or 1 so these R.V.s. are not taken into consideration.

→ Total distance b/w 2 disjoint R.V.s probability distributions is 1.

Let's take a to be the support of x .

$$\text{Probability } X \in [0,1] \rightarrow 1/2 + 1/2 = 1$$

$$X+a \in (0,1) = 0 \cdot 1 + 1 \cdot 1 = 1$$

So, the worst probability that the Total Variation between 2 probability Distributions is 1.

$$\therefore \left| \underbrace{P(X \in [0, 1])}_{=1} - \underbrace{P(X+2 \in [0, 1])}_{=0} \right| = 1.$$

If Two R.V.s have disjoint support then their TV is 1.

If 2 probability distributions have even 0 partial overlap:

Eg if $q=1$

$X=0$ w.p. $\frac{1}{2}$	$X \neq 1$ w.p. $\frac{1}{2}$
$X=1$ w.p. $\frac{1}{2}$	$X \neq 2$ w.p. $\frac{1}{2}$

$$= \frac{1}{2} \left(\frac{1}{2} + 0 + \frac{1}{2} \right) = \frac{1}{2},$$

$$\begin{array}{lcl} X=0 & X=op & = p - p - \frac{(1-p)}{1+p} \\ X=1-p & X=1-p & \\ 1 & 1 & \end{array}$$

CLT:

$$\begin{array}{l}
 X_1 \dots X_n \\
 \text{i.i.d Ber}(1/2)
 \end{array}
 \quad
 \frac{\sqrt{n} \left(\bar{X}_n - 1/2 \right)}{\sqrt{1/2(1-1/2)}}
 = 2\sqrt{n} \left(\overset{\text{Ber}(1/2)}{\bar{X}_n} - 1/2 \right)$$

$\downarrow d(n \rightarrow \infty)$
 $Z \sim N(0,1)$

Values taken by $2\sqrt{n}(\bar{X}_n - 1/2)$

$$\begin{array}{c}
 = 2\sqrt{n} \left(\underbrace{\left\{ 0, 1/2, \dots, \frac{1}{n} \right\}}_{\substack{\text{Size } n+1 \\ \text{discrete}}} - \frac{1}{2} \right) \\
 \text{Ber}(\bar{X}_n), \text{Ber}(1/2)
 \end{array}$$

S_n is the support of $\left[2\sqrt{n}(\bar{X}_n - 1/2) \right]$
 $\underbrace{\hspace{10em}}_{\text{Size } n+1}$

$\downarrow$
 we don't have a formula
 for discrete vs continuous.

$$\text{Ber} \left(\frac{1}{2}, \frac{1}{2} \right)$$

Things don't go well when TV is equal to 1.

And I have 1 probability which is equal to 1 and 1 which is equal too.

I just take $A = S_n \rightarrow (\text{Support})$

Defⁿ of support:

A belongs to $S_n(\text{support})$
with probability 1.

A Gaussian belongs to S_n (a discrete set), is probability 0.
($\therefore$ continuous).

$$\therefore [1-0] = 1 = TV.$$

$(1, 1, \dots, 1, \dots)$ which never is 0

↑
This is what is called trivial metric.

For a person $\rightarrow$ Same person $\rightarrow 0$
 $\downarrow$
Distance b/w 2 persons = 1

↓
Trivial Distance

- $\rightarrow$ It satisfies the Δ inequality
- $\rightarrow$ It is symmetric
- $\rightarrow$ It is definite,
- $\rightarrow$ It is +ve,

$\hookrightarrow$ Total variation is fine when things go well. But otherwise it captures nothing.

Total variation $\rightarrow$ Does not capture proximity.

Eg $\rightarrow X+a \rightarrow$ can go close 1. But Total variation does not capture it.

$\downarrow$

It's equal to 1, as long as $a \neq 0$.

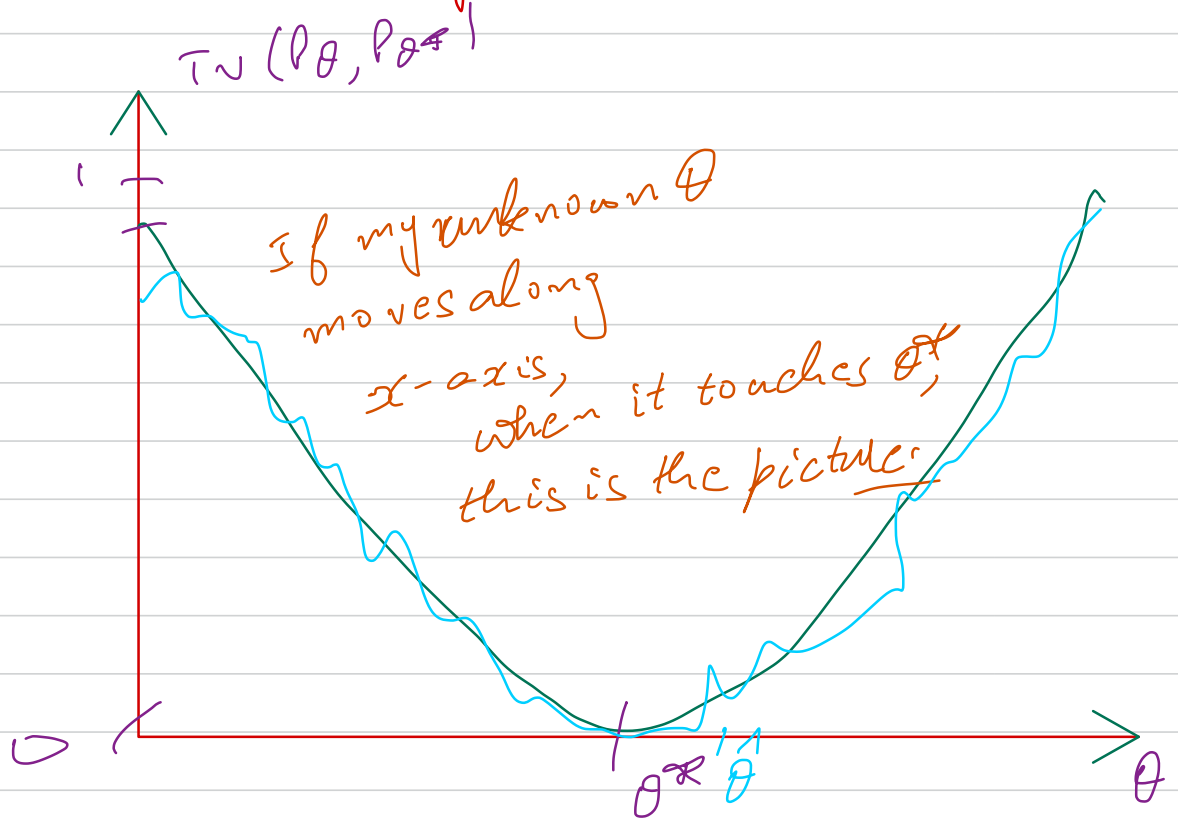
$\rightarrow$ It does not capture horizontal movements.

Eg $\rightarrow$ when I take my R.V., and move it a little bit, TV stops paying attention when they are disjoint (I think area under the curve)

$$\frac{1}{2} \left(1 + 1 - e^{-\rho} \right)$$

$$\frac{1}{2} \left(\rho - e^{\rho} \right)$$

KL-Divergence:



$\theta = \theta^*$, area under the curve $\rightarrow$

$\rightarrow$ if my estimator is good, my TV would be close θ^* , MINIMUM OF $TV = \theta^*$.

we can suggest $\hat{\theta}$, to be the estimator of θ^* ,

→ This $\hat{\theta}$ is the principle behind every estimator we are going to talk about.

→ If you have a fn. that is minimized at the parameter you are looking for, try to estimate that fn.

AND MINIMIZE THE ESTIMATOR
INSTEAD.

Unclear
how to build
an estimator
for total variation

It becomes trivial

↓ How?
Let's see methods for
estimation.

Then, I am saying, I need to estimate something.

↓
How do I do this?

↓
MAXIMUM LIKELIHOOD ESTIMATION
OF THE TOTAL VARIATION.

↓ what do you do!

consistent Replacing the avg.

↓
like we did for the KISS example.

↓
That's our building block.

(the LLN is a good guiding principle)

→ this fn if I can express as something that is minimized at θ^*

$$\Phi\left(\frac{x-\mu}{\sigma}\right) = 0.5$$

$$\sim P(X < \mu) = 0.5.$$

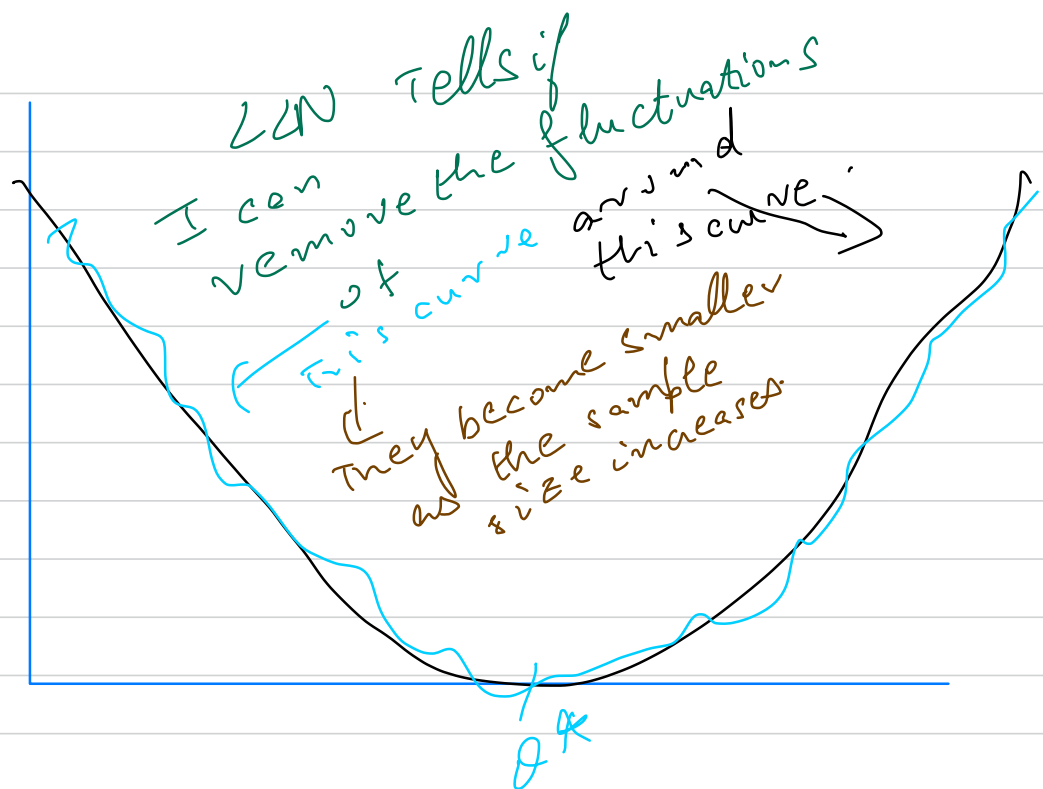
$$\Phi(\mu) = 0.5.$$

As the expectation of something

And you can replace the expectation by avg.

$$[0.5 - 1]$$

Now you have an estimator.



Replaces the TV, by another notion of distance between probability measures, but

Expectation of something.

I can say
 (let me replace
 the expectation
 by avg.)

Kullback-Leibler Divergence

→ Relative Entropy:

Twice
Asymmetric
once

Information
Theory

$$TV = \frac{1}{2} \int |f_{\theta^1}(x) - f_{\theta^2}(x)| dx$$

↓
L1-norm

Defⁿ of K-L divergence:

Let P and Q be discrete probability distributions with pmfs p and q , respectively.

$$P, Q \in \mathcal{E}.$$

K-L divergence (also known as relative entropy)

between P and Q is defined by .

$$KL(P, Q) = \sum_{x \in \mathcal{E}} p(x) \left(\ln(p(x)) - \ln(q(x)) \right)$$

$$a \ln(b) + (1-a) \ln(1-b)$$

$$= \sum_{x \in \mathcal{E}} p(x) \ln p(x) - \sum_{x \in \mathcal{E}} p(x) \ln q(x)$$

Sum is only the support of P ,

$$= \int_{x \in \mathcal{E}} p(x) \ln \left(\frac{p(x)}{q(x)} \right) dx$$

where this integral is only over the support of P .

Why only over P ?

Any point $x \in E$ outside support of q
where $q(x) \neq 0$.

$$\begin{aligned} \rightarrow \lim_{P/q \rightarrow 0^+} q \left(\frac{P}{N} \right) \ln \left(\frac{P}{q} \right) &= q \lim_{P/q \rightarrow 0^+} \left(\frac{P}{q} \right) \ln \left(\frac{P}{q} \right) \\ &= q \cdot (0) = 0. \end{aligned}$$

(By l'Hospital's Rule).

It is ^{not} a distance, it's a divergence.

Properties of KL-divergence:

→ (lots) (of lacks)

▷ $KL(P_\theta, P_{\theta'}) \neq KL(P_{\theta'}, P_\theta)$ in General.

▷ $KL(P_\theta, P_{\theta'}) \geq 0$.

▷ If $KL(P_\theta, P_{\theta'}) = 0$, then $(P_\theta = P_{\theta'})$ definite.

▷ $KL(P_\theta, P_{\theta'}) \neq KL(P_\theta, P_{\theta''}) + KL(P_{\theta''}, P_\theta)$
in General.

→ Attempts to make it symmetric
by summation over
 $p(x)$ and $q(x)$

Not getting
much apart from
symmetry.

Shannon-Jensen's divergence



Deep Neural Nets,

Symmetric $\rightarrow$ Nothing else.



$\rightarrow$ Jensen's Inequality $\rightarrow$ $\log$ is concave,
If applied to KL-divergence
It will show it is ≥ 0 .

$\rightarrow$ It is definite.

Definite:

INTUITION:

(The fact that it's only tells me it's at θ^*)

If those curves touch each other, any minimizer, I come up with could be one of those guys.

For T_0 , the only thing we used was the black curve would touch the x-axis was at θ^*

(Model Identifiable)

θ^* is the unique minimizer of $\theta \mapsto KL(p_{\theta^*}, p_{\theta})$.

(Even if it's not symmetric, it's still the unique minimizer) $\rightarrow$ identifiable models.

ESTIMATING THE K-L DIVERGENCE:

①

MAXIMUM LIKELIHOOD ESTIMATION

→ So, same estimation strategy for the new estimator.

↓
K-L.

Good notion $\therefore$ if I can find something that makes 0^* to 0

↑
Identifiability.

$$a / \ln\left(\frac{b}{1-b}\right) - \textcircled{a}$$

$$a / (\ln a - a) \ln(b).$$

Estimating KL divergence:

Need to find an
Estimator $\rightarrow$ MLE

$\rightarrow$

INTUITION:

$\rightarrow$ KL is a good notion

$\downarrow \rightarrow$
 $\therefore$ It finds something which makes it
equal to 0.

$\swarrow$
I know I found the right distribution
 $\downarrow$ (Identifiability)
(Right parameter)

STARTED WITH
THIS IDEA

KL divergence

$$KL(P_{\theta^*}, P_{\theta}) = E_{\theta^*}[\log P_{\theta^*}(x)] - E_{\theta^*}[\log P_{\theta}(x)]$$

why useful!

How?

$$KL(P_{\theta^*}, P_{\theta}) = \sum P_{\theta^*} \log \frac{P_{\theta^*}(x)}{P_{\theta}(x)}$$

$$= \sum P_{\theta^*} \log(P_{\theta^*}(x)) - \sum P_{\theta^*} \log(P_{\theta}(x))$$

$$= \sum P_{\theta^*} H_{P_{\theta^*}}(x) - \sum P_{\theta^*} H_{P_{\theta}}(x)$$

(definition of expectation)

$$= E_{P_{\theta^*}}[H_1] - E_{P_{\theta^*}}[H_2]$$

$$= E_{P_{\theta^*}}[\log P_{\theta^*}(x)] - E_{P_{\theta^*}}[\log P_{\theta}(x)]$$

constant.

Def:

$$E[g(x)] = \sum P(x) g(x)$$

$$= \underbrace{E_{p_{\theta^*}}[\log p_{\theta^*}]}_{\text{constant}} - E_{p_{\theta}}[\log p_{\theta}].$$

Remember: My one and only estimation strategy right now, is to replace expectations by averages.

$$= (\text{CONSTANT}_{p_{\theta^*}}) - E_{p_{\theta}}[\log p_{\theta}(x)].$$

(This would change) *(The value where it is minimized won't change)*

(I have data from this expectation.)
(I have data from p_{θ^*} .)

only trick:
LLN

Replace it with avg.

$$E_{p_{\theta^*}}[h(x)] \rightarrow \frac{1}{n} \sum_{i=1}^n (h(x_i)) \text{ by LLN}$$

$$\therefore KL(p_{\theta^*}, p_{\theta}) = \text{"constant"} - \frac{1}{n} \sum_{i=1}^n \log p_{\theta}(x_i)$$

Deriving the MLE:

$$\rightarrow \textcircled{1} \min_{\theta \in \Theta} KL(p_{\theta^*}, p_{\theta}) \Leftrightarrow \min_{\theta \in \Theta} -\frac{1}{n} \sum_{i=1}^n \log p_{\theta}(x_i)$$

(this means the θ that minimizes LHS, is the same as RHS)

(Just $\because$ I am shifting up and down the curve so whatever the argument was it just won't move)

$\rightarrow$ Just minimizing the negative

which I guess would be maximizing

(n^{th} or dev statistic)

$$\Leftrightarrow \max_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^n \log(x_i) -$$

$$\Leftrightarrow \max_{\theta \in \Theta} \frac{1}{n} \log \left(\prod_{i=1}^n p_{\theta}(x_i) \right)$$

log is an increasing fn.

$$\Leftrightarrow \max_{\theta \in \Theta} \prod_{i=1}^n p_{\theta}(x_i) -$$

minimized.

This is the maximum LIKELIHOOD PRINCIPLE.

Remark:

① The quantity $\hat{\theta}_n = \text{maximizer of } \prod_{i=1}^n p_{\theta}(x_i) -$ is referred to as the minimum likelihood estimator.

NOTE:

→ This is the same as the estimator $\hat{\theta}_n = \text{minimized of } - \frac{1}{n} \sum_{i=1}^n \log(p_{\theta}(x_i)) -$

Remark 2: Under certain technical conditions,
the maximum likelihood estimator is guaranteed
to converge to the true parameter θ^*

Likelihood Estimator:

→ A likelihood of a model,

→ Discrete $\&$ →

↳ It's actually a n -sample and a
candidate parameter.

Definition:

Likelihood of a model is defined as:

$$L_n: (\mathcal{E}^n \times \Theta \rightarrow \mathbb{R}^+) \rightarrow \text{map } L_n \text{ is just } L$$
$$(x_1, \dots, x_n, \theta) \rightarrow p_\theta[x_1=x_1, \dots, x_n=x_n]$$

Likelihood describes
can be described
for many
times

MEANING
OF LIKELIHOOD

→ How likely this data was
generated by p_θ

→ Probability
of likelihood.

Let's look at the probability that those numbers were coming from θ .

it is. $x_1 p + (1-p)$ (the bigger n)
 The bigger this number, the more likely it is.
 $x = x_1, x = x_2, \dots, x = x_n$
 $p + (1-p), p + (1-p), \dots, (1-p) + (1-p)$
 That's what likelihood means.

$E^m \times \theta \rightarrow R$ (for different values of θ)

→ 2 candidate values: $\theta = 0$ or $\theta = 1$.

→ Which one is the more likely to have generated this data.
 ↓ More θ .

Dual Vision of Statistics And Probability -

(I am just computing the probability of having seen these numbers for different values of θ)

(Fix $x_i \rightarrow$ compute probability) ↑ as a fix θ

SIMPLE WAY TO THINK

To values of $\theta \rightarrow$

$\theta \rightarrow 0$
 $\theta \rightarrow 1$

You could try which one is the more likely to have generated your data

Example: $\text{Ber}(p)$ - $\text{Ber}(p) = \{0, 1\}$

$\rightarrow$ either

$\text{Ber}(p) = \{0, 1\}$

when $p=0$

Now, n samples:

$0, 0, 0, 0, \dots$

True parameter

Question: what is the prob; $x_1=0, x_2=0, \dots, x_n=$

$\text{Ber}(0) = 1$

if $1, 1, 1, 1, \dots, p=0$

$\text{Ber}(0) = 0$